

Semester-wise Pattern for Students Admitted to B.E. Chemical Programme								
Year	First Semester			U	Second Semester			U
I	BITS	F103	Introduction to Engineering Design and Prototyping	4	BITS	F111	Thermodynamics	3
	BIO	F101	Introduction to Biological Sciences	3	BITS	F112	Technical Report Writing	2
	CHEM	F101	Fundamentals of Chemistry	3	CS	F111	Computer Programming	4
	MATH	F101	Multivariable Calculus	3	MATH	F113	Probability and Statistics	3
	PHY	F101	Oscillations and Waves	3	MATH	F102	Linear Algebra and Complex Variables	3
	BITS	F101	Social Conduct	1	EEE	F111	Electrical Sciences	3
	BITS	K101	Physical Well-being and Creativity	1*	BITS	F102	Innovation and Design Thinking	1
				18				19
II	MATH	F211	Mathematics III	3	ECON	F211	Principles of Economics	3
			Humanities Electives	3(min)			or	or
	CHE	F211	Chemical Process Calculations	3	MGTS	F211	Principles of Management	3
	CHE	F214	Engineering Chemistry	3			Humanities Electives	3(min)
	CHE	F213	Chemical Engineering Thermodynamics	3	CHE	F241	Heat Transfer	3
	CHE	F212	Fluid Mechanics	3	CHE	F242	Numerical Methods for Chemical Engineers	3
	BITS	F225	Environmental Studies	3	CHE	F243	Material Science & Engineering	3
				CHE	F244	Separation Processes I	3	
				21 (min)				18(min)
Summer BITS F221 Practice School – I(for PS Option Only)								
III			Open/Humanities Electives	3to6			Open/Humanities Electives	3to6
	CHE	F312	Chemical Engineering Laboratory I	3	CHE	F341	Chemical Engineering Laboratory II	3
	CHE	F313	Separation Processes II	3	CHE	F342	Process Dynamics & Control	3
	CHE	F311	Kinetics & Reactor Design	3	CHE	F343	Process Design Principles II	3
	CHE	F314	Process Design Principles I	3			Discipline Electives	6
			Discipline Electives	3				
				18/21				18/21
IV			Open Electives	5 to 11	BITS	F412	Practice School-II	20
			Discipline Electives	6			or	or
					BITS	F421T	Thesis or	16
					or	F425T	Thesis	18-20
					or	F424T	Thesis (9) and Electives (6 to 9)	15 to 18
				11/17				15/20

Discipline Core -45 Units (15 Courses)

Discipline Electives-15 Units (5 Courses)